

PID vs. Sliding-Mode Control for Net-Distance Keeping in Aquaculture Cage Inspection: Robustness to Marine Current Disturbances

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Abstract—Automated inspection of aquaculture net cages is essential for early detection of structural damage, yet reliable inspection depends on maintaining the underwater vehicle within a narrow distance range from the net. Recent work on deep-learning-based hole detection in fish net cages shows that detection confidence, based on YOLOv8, is strongly distance-dependent, remaining reliable within a 70–150 cm window and degrading significantly beyond 300 cm. However, the distance-keeping controller used in that system was validated only in still-water tank conditions, without current disturbances. This work investigates whether a proportional-integral-derivative (PID) controller can maintain this operational window under simulated marine current disturbances, and compares its performance against a Sliding Mode Controller (SMC), which is robust by design to bounded disturbances. Specifically, we address the following research question: up to what current disturbance does the PID controller fail to keep the vehicle within the desired window, and to what extent does SMC provide better performance? Results are evaluated through simulation in a ROS2 environment.

I. INTRODUCTION

Net inspection in aquaculture cages is a demanding, routine task: fish farms can require monitoring more than 1500 m² of net surface, traditionally carried out by human divers under high-risk, hostile underwater conditions. Undetected net damage can lead to fish escape, stock loss, and environmental harm, motivating growing interest in autonomous underwater vehicles (AUVs) and remotely operated vehicles (ROVs) equipped with computer vision for automated inspection.

A recent hybrid intervention system [2] addresses this problem using a BlueROV2 equipped with a CNN-based distance estimator and a YOLOv8 object detector to identify net holes, regulated by a proportional controller. Critically, hole-detection confidence is distance-dependent, remaining high within a 70–150 cm range and degrading significantly beyond 300 cm, making stable distance-keeping a necessary condition for reliable inspection. However, this controller was validated only in still-water tank conditions, without current disturbances, which are a persistent feature of open-water aquaculture sites.

This raises the question of whether a PID controller remains adequate under such conditions, or whether Sliding Mode Control (SMC), robust by design to bounded disturbances and already explored for underwater vehicle attitude tracking [3], is needed to preserve inspection reliability.

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II. SYSTEM MODELING

In order to build a ROS2 simulation, we need to derive a dynamic model of the BlueROV2 that is sufficiently accurate to reproduce the vehicle behavior during the net-distance-keeping task, while remaining lightweight enough for real-time integration in the ROS2 node used for comparing PID and Sliding Mode Control (SMC). The starting point is the general 6-DOF marine vehicle model proposed by Fossen [1], progressively simplified based on the physical characteristics of the BlueROV2 and the specific requirements of the inspection task, until obtaining a reduced 3-DOF model (surge, heave, yaw) used in all subsequent simulations.

A. Kinematics and Notation

Following standard marine robotics conventions, the vehicle motion is described with respect to an inertial North-East-Down (NED) frame and a body-fixed frame. The position/attitude vector and generalized velocity vector are:

$$\boldsymbol{\eta} = [x \ y \ z \ \phi \ \theta \ \psi]^T \quad \boldsymbol{\nu} = [u \ v \ w \ p \ q \ r]^T \quad (1)$$

The kinematic relationship between frames is:

$$\dot{\boldsymbol{\eta}} = J(\boldsymbol{\eta}) \boldsymbol{\nu}, \quad (2)$$

where $J(\boldsymbol{\eta})$ includes the rotation matrix R_b^n and the Euler-angle transformation matrix T_θ [1].

B. Full 6-DOF Dynamic Model

The general equation of motion for an underwater vehicle is:

$$M\dot{\boldsymbol{\nu}} + C(\boldsymbol{\nu})\boldsymbol{\nu} + D(\boldsymbol{\nu})\boldsymbol{\nu} + g(\boldsymbol{\eta}) = \boldsymbol{\tau}_{thrust} + \boldsymbol{\tau}_d \quad (3)$$

where each term represents respectively: inertial mass ($M = M_A + M_{RB}$), Coriolis/centripetal effects ($C(\boldsymbol{\nu})$), hydrodynamic

damping ($D(v)$), restoring forces (g), thruster-generated forces, and environmental disturbances [1].

C. Simplifying Assumptions for BlueROV2

For the net-distance-keeping task we adopt a 2D planar model (surge–sway–yaw), which is a common and well-validated simplification for compact ROVs operating at low speed. The following assumptions, consistent with the literature on BlueROV2 hydrodynamics [5], [6], allow reducing the full 6-DOF dynamics to a lightweight model suitable for real-time simulation:

- 1) Passive stability in roll and pitch. The center of buoyancy lies above the center of gravity, this creates a restoring moment that naturally drives roll and pitch back to zero without requiring active control. As a consequence, roll and pitch dynamics remain small during inspection maneuvers and can be safely neglected, reducing the model to the horizontal plane.
- 2) Negligible Coriolis terms at inspection speeds (< 0.5 m/s). $C(v) \approx 0$
- 3) Decoupled DOFs (diagonal M and D matrices). The BlueROV2 hull is nearly symmetric and the thruster layout is arranged to minimize cross-coupling between axes. This allows treating surge, sway and yaw as independent scalar subsystems, greatly simplifying controller design and stability analysis.
- 4) Near-neutral buoyancy ($g(\eta) \approx 0$ on controlled DOFs).

After applying the simplifications listed above, most hydrodynamic terms in Fossen’s model disappear.

$$M \dot{\nu} + D(\nu) \nu = \tau_{th} + \tau_d$$

with M and D diagonal matrices and:

$$\nu = [u, v, r]$$

As a result, the only hydrodynamic contribution that remains significant in the planar dynamics is the damping term $D(\nu)\nu$, which represents the drag generated by the relative motion between the vehicle and the surrounding fluid. It can be described by a linear part X_i proportional to ν and a quadratic one $X_{|i|}$ proportional to $|\nu|\nu$. Since the system is decoupled, each degree of freedom is affected only by its own drag component, leading to the following scalar expressions:

$$surge \rightarrow D_u(u_r) u = X_u u_r + X_{|u|u} |u_r| u_r$$

$$sway \rightarrow D_v(v_r) v = Y_v v_r + Y_{|v|v} |v_r| v_r$$

$$yaw \rightarrow D_r(r) r = N_r r + N_{|r|r} |r| r$$

Where:

$$X_{|u|u} = -\frac{1}{2} \rho C_D A_x \quad (4)$$

$$Y_{|v|v} = -\frac{1}{2} \rho C_D A_y \quad (5)$$

$$N_{|r|r} = -\frac{1}{2} \rho C_D A_n \quad (6)$$

These terms represent the quadratic hydrodynamic drag acting in surge, sway, and yaw, respectively, ensuring that the drag always opposes the motion. The parameters A_x , A_y , and A_n denote the cross-sectional areas exposed to the flow, ρ is the water density, and C_D is the drag coefficient.

In addition, the planar equations of motion include the inertial contribution associated with each degree of freedom. For underwater vehicles, inertia is not given solely by the rigid-body mass ($m_{RB,i}$): when the vehicle accelerates, it must also accelerate a surrounding volume of water, producing the so-called *added mass* ($m_{A,i}$). In Fossen’s formulation, the total inertia along each DOF is therefore the combination of rigid-body inertia and added mass. This leads to the following effective inertial terms:

$$I_u = m_{RB,u} + m_{A,u}, I_v = m_{RB,v} + m_{A,v}, I_z = m_{RB,r} + m_{A,r}. \quad (7)$$

These quantities appear in front of the acceleration terms $\dot{u}$, $\dot{v}$ and $\dot{r}$ in the final 3-DOF dynamic model and represent the total “effective mass” that the controller must accelerate in surge, sway and yaw.

D. Marine Current Disturbance Modeling

The effect of marine current enters the dynamics through the *relative velocity* between the vehicle and the surrounding fluid. Hydrodynamic damping depends on this relative velocity, not on the vehicle velocity alone. Therefore, the disturbance is modeled by subtracting the current velocity ν_c from the vehicle velocity ν :

$$\nu_r = \nu - \nu_c, \quad (8)$$

where ν_c is expressed in the body frame. Evaluating the damping terms on ν_r ensures consistency with Fossen’s formulation [1] and correctly captures the physical behavior: if the vehicle moves at the same speed and direction as the current, the relative velocity becomes zero and the drag vanishes.

In the planar case, only surge and sway are affected by the current, leading to:

$$u_r = u - u_c, \quad v_r = v - v_c. \quad (9)$$

E. 3-DOF Dynamic Model

Combining the effective inertial terms with the hydrodynamic damping evaluated on the relative velocity yields the final 3-DOF planar model used in simulation. Each DOF is decoupled and governed by a scalar nonlinear equation:

$$I_u \dot{u} = \tau_u + \tau_d + X_u u_r + X_{|u|u} |u_r| u_r, \quad (10)$$

$$I_v \dot{v} = \tau_v + \tau_d + Y_v v_r + Y_{|v|v} |v_r| v_r, \quad (11)$$

$$I_z \dot{r} = \tau_r + \tau_d + N_r r + N_{|r|r} |r| r. \quad (12)$$

This model is lightweight, decoupled and suitable for real-time simulation in ROS2, while retaining the dominant nonlinear hydrodynamic effects relevant for the net-distance-keeping task.

F. Hydrodynamic Parameters

DA COMPLETARE Table I summarizes the parameters used, based on the identification study [7] and validated datasets [5].

TABLE I
HYDRODYNAMIC AND RIGID-BODY PARAMETERS OF THE BLUEROV2
(FROM [5], [7]).

Symbol	Description	Value	Unit
$m_{RB,u/v}$	Rigid-body mass	11.5	kg
$m_{RB,r}$	Yaw inertia	0.16	kg·m ²
$m_{A,u}$	Surge added mass	-5.50	kg
$m_{A,v}$	Sway added mass	-12.70	kg
$m_{A,r}$	Yaw added mass	-0.12	kg·m ² /rad
X_u	Linear surge damping	-4.03	N·s/m
Y_v	Linear sway damping	-6.22	N·s/m
N_r	Linear yaw damping	-0.07	N·s/rad
$X_{ u u}$	Quadratic surge damping	-18.18	N·s ² /m ²
$Y_{ v v}$	Quadratic sway damping	-21.66	N·s ² /m ²
$N_{ r r}$	Quadratic yaw damping	-1.55	N·s ² /rad ²

III. CONTROL METHODS

A. PID Controller

The Proportional–Integral–Derivative (PID) controller regulates the distance error $e(t) = d_{\text{ref}} - d(t)$ by combining three corrective actions:

$$\tau(t) = K_p e(t) + K_i \int_0^t e(\sigma) d\sigma + K_d \dot{e}(t). \quad (13)$$

The proportional term $K_p e(t)$ produces an immediate reaction to the error: a larger deviation from the desired distance generates a stronger corrective force. The integral term accumulates the error over time and compensates steady-state offsets, which typically arise when the vehicle is pushed by a constant current. The derivative term $K_d \dot{e}(t)$ anticipates the error trend: if the vehicle is approaching the net too quickly, the derivative action slows it down before overshooting. Together, these three contributions allow the PID to stabilize the surge motion around the desired distance. However, being a linear controller, its robustness is limited: under persistent disturbances such as marine currents, the PID may require high gains or may exhibit oscillations and residual steady-state error.

B. Sliding Mode Control (SMC)

Sliding Mode Control is a nonlinear technique designed to guarantee robustness against bounded disturbances and model uncertainties. The controller defines a sliding surface that encodes the desired error dynamics:

$$s(t) = \dot{e}(t) + \lambda e(t), \quad (14)$$

where $\lambda > 0$ determines how quickly the error should decay. When $s(t) = 0$, the error follows the stable first-order dynamics $\dot{e} = -\lambda e$, ensuring convergence to zero.

The SMC input is composed of an equivalent control τ_{eq} , which compensates the nominal model, and a discontinuous

term that drives the system toward the sliding surface, accounting for disturbances :

$$\tau(t) = \tau_{\text{eq}} - k \text{sat}\left(\frac{s}{\phi}\right), \quad (15)$$

where $k > 0$ is the switching gain and $\text{sat}(\cdot)$ is a saturation function used to reduce chattering (oscillations at high frequency on s surface). The switching term actively counteracts disturbances: as long as the disturbance is bounded, the controller maintains $s(t) \approx 0$, making the closed-loop behavior insensitive to variations in hydrodynamic parameters and to external forces such as marine currents. This property makes SMC particularly suitable for maintaining the vehicle within the required distance window under challenging flow conditions.

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